full-mhs-z: implementation

Technical notes on `run.mode = "full-mhs-z"` (the M5 joint Gauss–Newton driver, M5i defaults). HSEMHS, 24 Oct 2026.

1. Running it

```
[mpiexec -n R] julia --project=. -t T scripts/run.jl config/modes/full-mhs-z.toml
```

`scripts/run.jl` validates the config, exports the M5 switches as environment variables (`MHS_INPUT`, `MHS_EOS_TABLE`, `MHS_SIGMA_REL`, `MHS_TRUST`, `MHS_STOP`, `MHS_PHI_RTOL`; they are read as constants when the driver is included), includes `scripts/m5_bifrost/m5h_fullres.jl` and calls `main([output, run, wall, ckpt, init_iter, restart])`. This is why the mode cannot go through `run_config`.

Two entry points, two sets of defaults. Through `run.jl` (this mode) the defaults are the M5i choice below. Run directly (`julia -project=. scripts/m5_bifrost/m5h_fullres.jl OUTDIR run ...`), the driver reads the same environment variables with its own defaults: 5 % relative σ, trust region on, height stop (`dz`): the M5h configuration. The environment constants are read once, at include time.

Forwarded keys. Only `input.file`, `eos.table_file`, `run.output` (a directory) and `full_mhs.*`. Not forwarded: `physics.g` (gravity from the file attribute), `[column_hse]`, `[mhs]`, `[parallel]` (the process grid is chosen by `bif_problem`), `[tau]`, `[output]`, `run.overwrite`, the input dataset names. $P_{\rm ref}$ comes from the attribute `P_ref_Pa`.

Key	Default (M5i)	Meaning
`input.file`	required	height-grid cube (production `IO/input-mhs-hse-fullz.nc`, $504 \times 504 \times 496$)
`run.output`	required	run directory
`full_mhs.run`	`run`	`run`: native grid; `smoke`: 42×42, every second level, every code path in minutes
`full_mhs.wall_hours`, `ckpt_hours`	24, 1	wall cap; checkpoint interval
`full_mhs.init_iter`	3000	CG iterations of the field initialisation
`full_mhs.restart`	`""`	any saved file (`init.h5`, `atmosphere_A/B.h5`, final)
`full_mhs.sigma_rel`	0	$\sigma_B = \sigma_{\rm rel}\lvert\mathbf{B}_{\rm obs}\rvert \oplus 5\,$G (M5h: 0.05)
`full_mhs.trust_region`	false	false: line search; true: trust region (M5h)
`full_mhs.stop`, `phi_rtol`	`phi`, 10^{-6}	$\lvert\Phi_{\rm prev} - \Phi_{\rm new}\rvert/\Phi_{\rm new} <$ rtol on 3 consecutive accepted steps (current policy); `dz` ($\max\lvert\Delta z_{\tau 1}\rvert < 1\,$mm) is a retained historical option

2. Problem setup (`bif_setup.jl`, `bif_problem`)

Reads T, $\mathbf{B}_{\rm obs}$, coordinates and the attributes `P_ref_Pa`, `period_x_m`, `period_y_m`, `gravity_m_s2`; `init_parallel` (2-D blocks); periodic `Grid`; σ per component from `sigma_synth`; `SMHSProblem` with $\lambda_F = 10^{-4}$, $\lambda_Q = 10^{-3}$, $\lambda_S = 10^{-3}$, $\lambda_U = 0$, data on faces. State $x = (v, \tilde{A}_x, \tilde{A}_y, \tilde{A}_z, \bar{b})$: $u = v - \langle v\rangle_{\rm bottom} + \ln P_{\rm ref}$ at centres (the gauge as a parameterisation), $\mathbf{A} = A_{\rm ref}\tilde{\mathbf{A}}$ on edges ($A_{\rm ref}$ = median σ × mean spacing), $\mathbf{B} = \nabla\times\mathbf{A} + \bar{b}\hat{\mathbf{z}}$ on faces, $\bar{b}$ a scalar stored on rank 0.

- `init_state`: `staggered_hse!`, the exact discrete vertical balance $(P_k - P_{k+1})/\Delta z_{k+1/2} + g\sqrt{\rho_k\rho_{k+1}} = 0$ upward from $P_{\rm ref}$ (Brent per level); $\bar{b} = \langle B_{z,\rm obs}\rangle$.
- `set_scales!`: $S = \max(\rho_0 g,\ 0.01\,\mathrm{median}_k\rho_0 g)$ at centres, $S_f^2 = S_a S_b$ on faces; Φ_Q layer weights $(B_{\rm ref}/S)^2$ from layer medians. Fixed for the solve.
- `init_A!`: $(\tilde{\mathbf{A}}, \bar{b})$ fitted to $\mathbf{B}_{\rm obs}$ with v frozen ($\Phi_B + \Phi_S$ only, quadratic): CG on the normal equations with $Qp = \nabla\Phi(p) - \nabla\Phi(0)$; converges to the minimum-norm solution in the curl gauge null space.
- Pressure floor `ulo` $= u_0 - 3$.

3. Objective (`MHSStaggered.jl`, `objective!`, `gradient!`)

$\Phi = \Phi_F + \Phi_B + \Phi_S + \Phi_Q$ (+ optional Φ_U, Φ_N, off): $\Phi_F = (\lambda_F/N_F)\sum_f w_f|R_f|^2/S_f^2$ on x- and y-faces of layers $2\ldots N_z - 1$ and the interior z-faces; $\Phi_B = (1/N_B)\sum((B_f - B_{{\rm obs},f})/\sigma_f)^2$ on faces ($\mathbf{B}_{\rm obs}$ and $1/\sigma^2$ interpolated to faces); $\Phi_S = (\lambda_S/N)\sum w\sum_a(h_a^2\partial_a^2\delta B)^2/\sigma^2$ at centres; $\Phi_Q = (\lambda_F\lambda_Q/N_F)\sum_e w_e\sum_a(B_{\rm ref}\frac{1}{4}\delta_a^2 J_e/S_e)^2$ on edges. On an x-face the force uses $\partial_x P$ from the two centres, J_y, J_z averaged from the nearest edges, B_y, B_z from the four nearest faces; on z-faces $\rho = \sqrt{\rho_k\rho_{k+1}}$. The gradient is a hand-written adjoint of every stencil, including the EOS term $\partial\ln\rho/\partial\ln P|_T$ from the table.

4. Solver (`MHSJointGN.jl`, `joint_gn_solve!`)

One quadratic model per iteration at the accepted base state:

1. PCG (Steihaug–Toint) on $(H_{\mathrm{GN}} + \mu D)\, y = -g$, H_{GN} products by `gn_hvp!` (linearised forward plus adjoint), D the preconditioner diagonal (`gn_diagonal!`: exact GN diagonal for v, per-layer Hutchinson estimates for $\tilde{\mathbf{A}}$), multiplied by `block_scale` $= (100, 1, 1)$ for $(v, \tilde{\mathbf{A}}, \bar{b})$; pressure nodes at the floor with an outward gradient fixed; at most 100 iterations, relative residual 10^{-2}.
2. Candidate: the pressure part re-centred on the anchor and projected onto the box $\max(u_{\mathrm{lo}} - u, -5) \le \delta u \le 5$; the field part unconstrained.
3. Projected-Cauchy safeguard along $-D^{-1}g$ (exact model minimiser without a trust region); used if the GN step achieves less than 0.1 of the Cauchy model decrease.
4. Trial in a separate workspace. M5i (`trust_region = false`): accept if Φ decreases (actual/predicted $> 10^{-4}$, floor respected), otherwise halve along the same direction (up to 30 times); μ fixed at 10^{-6} (gauge null space only). M5h: trust radius in the D-norm, μ and Δ adapted from the ratio.

The curl-free (gauge) directions of $\tilde{\mathbf{A}}$ carry no curvature; μD keeps the system definite and gauge components change neither $\mathbf{B}$ nor Φ.

5. Driver (`m5h_fullres.jl`)

After every accepted step (callback): $z(\tau_{500}{=}1)$ in every column from the current P (`tau1_heights!`, `exp_tail`), a row in `history.csv` (Φ and its terms, $dz_{\max}$, force and data statistics, step data), the stop test, checkpoints every `ckpt_hours` alternating `atmosphere_A/B.h5` (written to `.tmp`, then renamed), a `STOP` file check and the wall cap. Every candidate goes to `candidates.csv`. Saved solver arrays are global owned values (no halos), so a restart may use another rank count; on restart the grid is checked and the $z(\tau{=}1)$ map is recomputed and compared with the saved one (flags exact, heights to 10^{-6} m). That check does *not* establish that input, EOS table, σ or solver settings are the same as in the saved run; a restart with changed settings is a new experiment. Any error writes `atmosphere_crash.h5` (single rank) or aborts all ranks. `full_mhs.run = "run"` requires the 504×504 horizontal grid of the production cube.

Output directory. The driver creates `run.output` if needed and writes into it whether or not it exists (`run.overwrite` is not consulted); existing `history.csv`, checkpoints and final files are overwritten. Use a fresh directory per run, or a deliberate restart.

Termination and status. The solver returns a status and reason, logged as `solver returned: <status> (<reason>)`: the Φ criterion met (`callback stop` with the reason "user criterion met"), wall cap, `STOP` file, or line-search failure (30 halvings). `scripts/run.jl` exits 0 after the driver returns in all of these cases (code follow-up: propagate the status); only an exception gives a non-zero exit. A met Φ criterion does not by itself certify small force residuals or an acceptable data fit: read Φ_F, Φ_B and the residual and data statistics in `history.csv`.

6. Parallelisation

2-D (x, y) blocks with one halo layer (`exchange_halos!`), `THREAD_FUNNELED`, threads inside each block. Global sums use a fixed reduction order (row partials, `reduce_rows`), so gradients and objectives are reproducible; tests check thread and rank invariance of the staggered gradient. Output is cooperative (rank 0 creates the file, then each rank writes its hyperslab in turn); no parallel HDF5. Memory: the setup (`bif_setup.jl`) reads the *complete* raw input arrays (T and three field components) on every rank before filling the distributed fields, so each rank needs that memory in addition to its share of the solver arrays. Multi-node: MPICH via `mpiexecjl` with the ssh launcher, each rank wrapped in `mhs_rank.sh` (libfabric TCP on the IPoIB interface, node-local Julia), see `config/modes/README.md`.

7. Verification and performance

- Test suite: staggered gradient against finite differences (ideal gas, Wittmann table), forward/adjoint allocations, thread and rank invariance, the exact discrete HSE initialiser, Gate C1b (uniform $\mathbf{B}$), C2 (potential field), C3 (checkerboard modes), and the M5–M5g solver tests (projection, block termination, trust ball, barrier and face bounds).
- Through `run.jl`: smoke run on 4 ranks ($42 \times 42 \times 249$) completed 3 accepted steps, checkpoints and the final save; the log confirmed $\sigma_B = 5$ G uniform, no trust region, Φ stop at 10^{-6}.
- Production M5i (22–23 Oct 2026, branch `m5i-phi-notrust` commit c96a922, run directory `out/m5i/run_fullz` in the cluster worktree `MHS_m5i`): 24 ranks × 16 threads on 4 nodes, $\approx 5 \times 10^8$ unknowns; 112 accepted steps, $\Phi = 7.88 \times 10^3$ when stopped on request. M5h (trust region, 5 % σ) ran to its 24 h wall cap. Neither met its stopping rule; these are progress records, not converged equilibria.